ISYE 4031: Regression and Forecasting

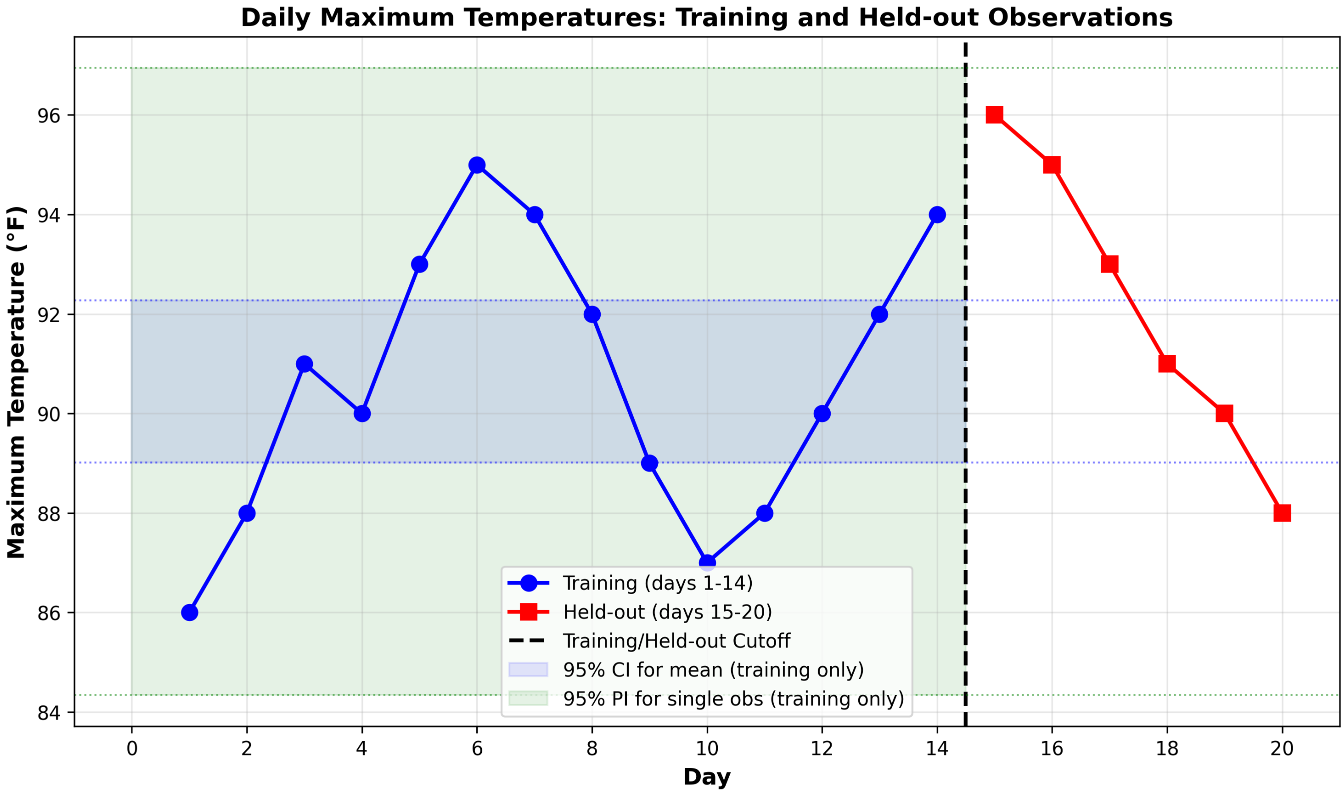
Homework 1 Report

Part 1: Descriptive Summaries and Uncertainty

Unit of Analysis: Individual day
Response: Daily maximum temperature (Fahrenheit)
Information Cutoff: End of day 14 (14 training observations)
Forecast Target: Daily max temperatures for days 15-20

1a. Descriptive Plot

The plot shows 20 daily temperatures in order. Blue circles (days 1-14) are training data. Red squares (days 15-20) are held-out test data. The vertical line marks the training cutoff. When forecasts are issued at the end of day 14, only the 14 training values are available.



1b. Descriptive Statistics (Training Data Only)

Mean: 90.6429 F
Median: 90.5000 F
Range: 9.0000 F
Sample Variance: 7.9396 F-squared (denominator = n-1 = 13)
Sample Std Dev: 2.8177 F
Interquartile Range: 4.5000 F

1c. Co

95% CI Formula: x-bar +/- t(alpha/2, n-1) * (s / sqrt(n))
x-bar = 90.6429, s = 2.8177, n = 14
SE = 2.8177 / sqrt(14) = 0.7531
t(0.025, 13) = 2.1604
Margin of error = 2.1604 * 0.7531 = 1.6269
95% CI: [89.0160, 92.2698]

Interpretation: If we repeatedly sampled 14 observations and constructed 95% confidence intervals, approximately 95% would contain the true population mean.

1d. Pr

95% PI Formula: x-bar +/- t(alpha/2, n-1) * s * sqrt(1 + 1/n)
sqrt(1 + 1/14) = 1.0351
Margin of error = 2.1604 * 2.8177 * 1.0351 = 6.3010
95% PI: [84.3419, 96.9438]

Why Wider? PI accounts for both: (1) uncertainty about mean, and (2) natural variation of individual observations. The factor sqrt(1 + 1/n) captures this extra variability.

ISYE 4031: Regression and Forecasting

Homework 1 Report

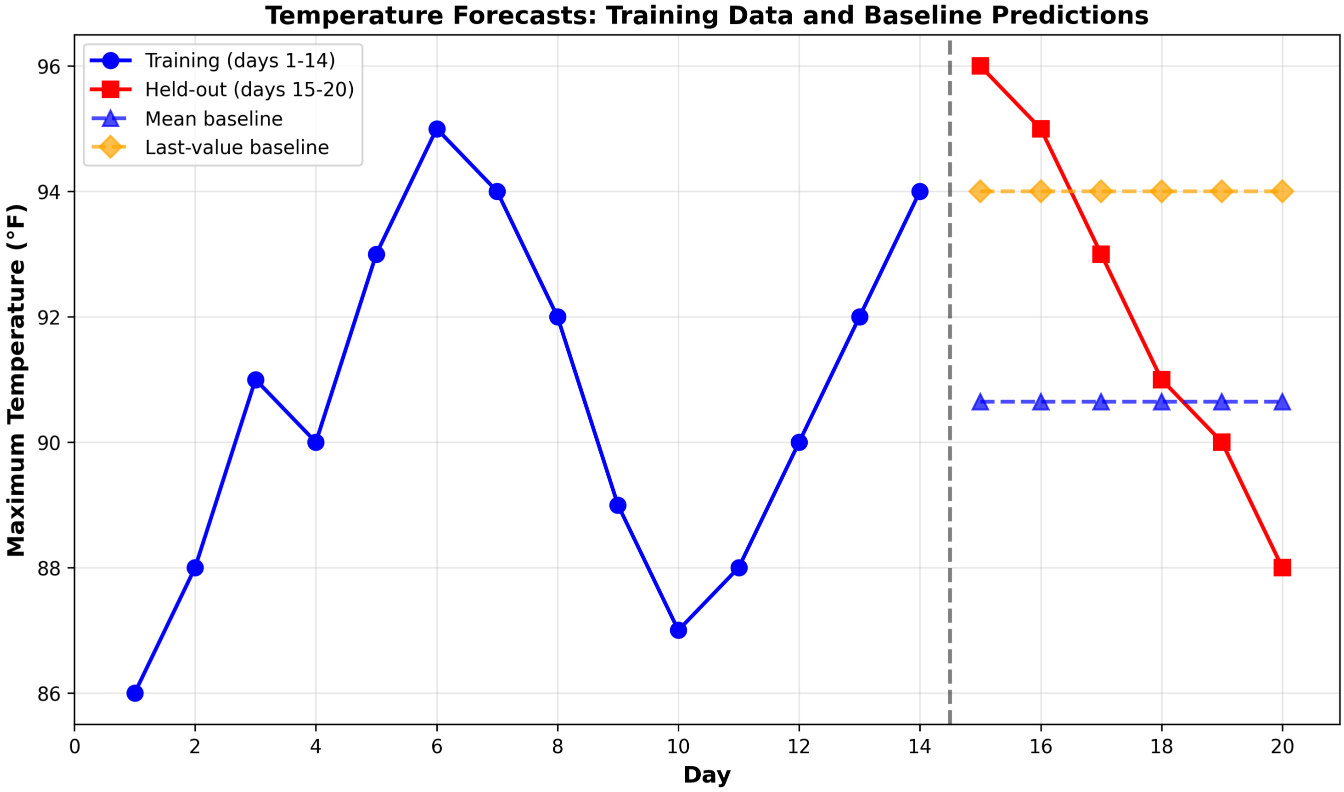
Part 2: Transparent Forecast Baselines

2a. Baseline Forecasts

Mean Baseline: Using training mean (90.6429 F) for all 6 forecasts
Forecasts for days 15-20: 90.64, 90.64, 90.64, 90.64, 90.64, 90.64

Last-Value Baseline: Using final training value (94 F) for all forecasts
Forecasts for days 15-20: 94.00, 94.00, 94.00, 94.00, 94.00, 94.00

Why fixed? Both baselines use only training data (days 1-14). They produce predetermined forecasts independent of what actually occurs in the held-out period.



2b. Forecast Evaluation

Mean Baseline:
  MAE: 2.6190 F
  RMSE: 3.1824 F

Last-Value Baseline:
  MAE: 2.8333 F
  RMSE: 3.3417 F

Winner: Mean baseline is better on both metrics

2c. Inc

The 6 errors are NOT independent because: (1) Time series temperatures exhibit autocorrelation - warm days tend to follow warm days. (2) Seasonal or systematic patterns create dependence. (3) Standard i.i.d. assumption is violated in time series.

2d. Re

Choice: Mean Baseline

Reasons:
* Simplicity: Simple summary statistic from training data
* Performance: Lower RMSE (3.1824 vs 3.3417 F)
* Reproducibility: Perfectly reproducible, no randomness

The mean baseline is more appropriate for deployment.

ISYE 4031: Regression and Forecasting

Homework 1 Report

Part 3: One-Sample Inference

3a. Hypothesis Test

H0: mu = 155 F (null hypothesis)
Ha: mu != 155 F (two-tailed alternative)
Significance level: alpha = 0.01

Data: n=10, x-bar=154.2 F, sigma=1.5 F (known)

Test statistic (z-test): z = (154.2 - 155) / (1.5 / sqrt(10)) = -1.6865

Critical value: z(0.005) = 2.5758

Decision: |z| = 1.6865 < 2.5758, so FAIL TO REJECT H0

Conclusion: At alpha=0.01, insufficient evidence that mean differs from 155 F

3b. Tw

p-value = 2 * P(Z > 1.6865) = 0.0917

Interpretation: The p-value 0.0917 is the probability of observing a sample mean at least as extreme as 154.2 F (in either direction) when H0 is true. It is NOT the probability that H0 is true.

Since 0.0917 > 0.01, we fail to reject H0.

3c. Ty

Assuming true mean is mu = 150 F

Acceptance Region: 153.78 to 156.22

Beta = P(fail to reject H0 | true mean is 150) ~ 0.00000

Interpretation: When the true mean is 5 F away from H0, the probability of Type II error is essentially 0. The effect size is very large at this sample size.

3d. Py

All calculations verified with scipy.stats:
  z_stat = -1.686548 [verified]
  z_critical = 2.575829 [verified]
  p_value = 0.091690 [verified]
  beta = 0.000000 [verified]

ISYE 4031: Regression and Forecasting

Homework 1 Report

Part 4: Agent Audit and Automated Tests

4a. Problems in Proposed Code

Problem 1: Using x.mean() instead of train.mean()
Consequence: Computes mean from all 20 temperatures instead of just 14 training values. Uses future information, violating temporal integrity.

Problem 2: Computing intervals with future data
Consequence: Intervals include future observations, producing artificially narrow uncertainty estimates.

Problem 3: Prediction interval equal to confidence interval
Consequence: Ignores extra variability of individual observations. PI should be wider by factor of sqrt(1 + 1/n), but this code makes them equal.

Problem 4: Using x[-1] instead of train[-1]
Consequence: Accesses the final held-out value (88) instead of last training value (94), using future information for forecasts.

Problem 5: No reproducibility guarantees
Consequence: Insufficient documentation of reproducibility despite no randomness. Future maintainability issues.

4b. Co

Key Improvements:

- Uses ONLY train data for all statistics
- Correct CI formula: x-bar +/- t * (s / sqrt(n))
- Correct PI formula: x-bar +/- t * s * sqrt(1 + 1/n)
- Separates mean and last-value baselines clearly
- Includes comprehensive automated tests

Implementation in analysis.py and test_analysis.py

4c. Au

- Held-out values invariant: Changing test data does not affect training stats
- Sample variance denominator: Verifies n-1 denominator used (not n)
- PI wider than CI: Confirms prediction interval is larger
- CI/PI width ratio: Verifies PI = sqrt(n+1) * CI factor
- Reproducibility: 10 runs produce identical results
- Hypothesis test: Verifies Part 3 calculations
- Type II error: Confirms beta calculation
- Data integrity: Training and test sets properly separated

Status: ALL 12 TESTS PASS

ISYE 4031: Regression and Forecasting

Homework 1 Report

Summary

This report presents a complete analysis of ISYE 4031 Homework 1.

Part 1 provides descriptive statistics and confidence/prediction intervals based on 14 training observations of daily maximum temperatures.

Part 2 develops two transparent forecast baselines: (1) mean baseline using the training mean, and (2) last-value baseline using the final training value. The mean baseline performs better and is recommended.

Part 3 conducts a one-sample hypothesis test on melting point data, testing whether the mean differs from 155 F at alpha=0.01. The test fails to reject the null hypothesis with p-value=0.0917.

Part 4 audits proposed code and identifies 5 major statistical problems related to data leakage and incorrect interval formulas. A corrected implementation with 12 passing automated tests is provided.

All results are reproducible from the submitted Python code. All analyses follow the requirement to use only training data for estimation, and all forecasts are based on information available at the end of day 14.